

Pankaj Kumar

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Education

B.Tech in Information Technology — IEST, Shibpur

Aug 2023 – May 2027

CGPA: 7.77 / 10 · Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Achievements

- Ranked **218** among **22,000+** participants in CodeChef Starters 245
- Maximum LeetCode Rating :**1605** (Top 23%) · Solved **320+** DSA problems

Experience

Machine Learning Intern — Tamizhan Skills (RISE Program)

Jun 2026 – Jul 2026 · Remote

- Developed and evaluated **classification** and **clustering** models using Scikit-learn, applying end-to-end ML workflows on real-world datasets.
- Performed data preprocessing, feature engineering, and model validation, improving predictive performance through systematic experimentation and evaluation.

Projects

DrawChain — Real-Time Multiplayer Drawing & Guessing Game

[Live Demo](#) | [GitHub](#)

Next.js · TypeScript · Socket.IO · Redis · PostgreSQL · Prisma

- Architected a real-time multiplayer game engine using Socket.IO and Redis, enabling synchronized canvas updates with **sub-100ms latency** across concurrent players.
- Reduced client asset payload size by **90%** (**6.5MB** → **675KB**), significantly improving page load and rendering performance.
- Implemented secure session-based authentication and automated leaderboard persistence using NextAuth, Prisma, and PostgreSQL.
- Designed event-driven gameplay workflows for drawing synchronization, round management, scoring, and real-time player interactions.

IPL Win Predictor — Real-Time Match Outcome Prediction

[Live Demo](#) | [GitHub](#)

Python · Pandas · Scikit-learn · XGBoost · SHAP · Streamlit

- Engineered an automated data pipeline to process and normalize data from **1,200+ IPL match files**, generating a structured dataset for predictive modeling.
- Designed cricket-specific features including Required Run Rate, Wickets Remaining, and a custom Pressure Index; trained XGBoost models achieving **75% accuracy** on unseen seasons.
- Deployed an explainable prediction platform using Streamlit and SHAP, providing real-time win probabilities and model interpretation.

YourBooking — Hotel Booking Web App

[Live Demo](#) | [GitHub](#)

MongoDB · Express.js · React.js · Node.js

- Built a full-stack hotel booking platform with secure authentication, role-based access control, and reservation management workflows.
- Designed RESTful backend services and MongoDB data models supporting booking, user, and administrative operations.

Technical Skills

Languages: C++, Python, JavaScript, TypeScript, SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Frameworks & Libraries: React.js, Next.js, Express.js, Scikit-learn, XGBoost

Backend & Databases: Node.js, REST APIs, Socket.IO, PostgreSQL, MongoDB, Redis, Prisma ORM

Machine Learning: Pandas, NumPy, Feature Engineering, Model Evaluation, SHAP

Tools: Git, GitHub, Postman, Streamlit

Certifications

Artificial Intelligence Fundamentals — IBM · Credly Verified Badge

Covered Machine Learning, Neural Networks, and AI tools using Python and IBM Watson.